United International **University** (UIU)

Dept. of Computer Science & Engineering (CSE)

COURSE OUTLINE

Spring 2026

Course Code/Title: **CSE 6179 – Digital Forensics** Section: **M**

Instructor **Mohammad Mamun Elahi**, Assistant Professor, Dept. of CSE
Director, Center for IT Services (CITS)
LMC & Curriculum Head, Cisco Networking Academy.
mmelahi@cse.uiu.ac.bd, 01914-001570

Classes Lectures: **Sat @ 11 am** (Sec – M) at **Room # 0424** (**Computer Lab 20**)

Counseling hr **Sat:** 9:00 am to 11:00 am.

Room No. **518 (CITS)** **Official Contact No. +88 09604-848-848**-Ext.**1401**

Text Book **Digital Forensics,** AndrÈ Årnes, Wiley, 2018. **(DF)**

CT/Quiz Total **4 class tests** will be held. The **best 3** will count.

Exam Policy **No Mid-term** exam, **final exam** will be taken physically **in-person on-campus**.

Grading The course grade will be determined from a weighted average of the Class tests, homework assignments, term Project and a research/survey paper with presentation is as follows:

Attendance	5%
Class Tests	20%
Home works / Assignments	5%
Term Project (Individual)	15%
Survey Paper + Presentation (Group-wise)	15%
Final Exam	40%

Course Grade The following scale will be used to convert numerical grades to letter grade:

Grade	Marks	Grade Point	Grade	Marks	Grade Point	Grade	Marks	Grade Point	Grade	Marks	Grade Point
A (Plain)	90-100	4.0	B (Plain)	78-81	3.00	C (Plain)	66-69	2.00	D (Plain)	55-57	1.00
A- (Minus)	86-89	3.67	B- (Minus)	74-77	2.67	C- (Minus)	62-65	1.67			
B+ (Plus)	82-85	3.33	C+ (Plus)	70-73	2.33	D+ (Plus)	58-61	1.33			

Objectives This course covers both the fundamental knowledge and hands-on practice on computer and network forensics. Upon completing this course, the students are expected to understand the basics of computer and network forensics, to be well-trained as next-generation computer crime investigators, and to be prepared for active research at the forefront of these areas. A survey on state-of-the-art research opportunities will be explored in the area of cyber forensic investigation to give students a wider scope of knowledge. Students will work in groups to review/survey literature on a specific research area to demonstrate their understanding of contemporary issues and challenges in digital forensic investigation.

Outline The topics to be covered include:

Lecture #	Topics	Readings
1 (28 Feb)	Introduction to Digital Forensics	Chapter 1 (DF)
2 (7 Mar)	Forensic Investigation Process	Chapter 2 (DF)
3 – 4 (14 & 28 Mar)	Computer Forensics	Chapter 5 (DF)
5 – 6 (4 & 11 Apr)	Network & Internet Forensics	Chapter 7 (DF)
	Survey Paper Proposal	
7 (2 May)	Cyber Law – Complying with Legal & Industry	Chapter 3 (DF); Handouts
8 (9 May)	Memory Forensics – Memory & Malware Analysis	Papers/Articles
9 (16 May)	Cybercrime, Digital Evidence & Incident Management	Papers/Articles
10 (23 May)	Connecting the Dots – Event Logs	Papers/Articles
11 & 12 (6 & 13 June)	**Project Evaluation & Survey Paper Presentation**	